

LESSON 1: MACHINE LEARNING

One of the fundamental attributes of an intelligent system is **learning**—the ability to acquire and use information. Learning involves creating internal links, or **associations**, between what is already known and new information. At its core, learning is the capacity to form new associations and to make use of them.

LEARNING IS THE ABILITY TO FORM AND USE NEW ASSOCIATIONS.

EXAMPLE 1.1

Consider a toddler. Through experience, she recognizes the shape of a cup and understands that it contains water to quench her thirst. These are associations already present in her mind: <cup as an object>, <water>, <thirst>. She learned them through repeated experience.

Now something new happens. Maya feels thirsty and points to the cup. Her mother says, "cup" and hands Maya the cup. The next day, the same event repeats: Maya points to the cup, her mother says, "cup," and gives it to her. After a few days of this repeating pattern, Maya feels thirsty but does not see the cup nearby. This time, she looks at her mother and says, "cup."



Through repetition, Maya recognized a pattern: whenever she heard the word 'cup', she received the cup itself. This formed an association in her mind between the word 'cup' and the object. She learned that the object's name is 'cup'.

Maya has learned. She created a new association in her mind: linking the object cup to the word ‘cup’. This learning occurred because she recognized a repeating pattern—the sound "cup" was always paired with the actual cup—and reinforced it through experience.

From this simple example, we can identify three fundamental attributes of learning:

1. It requires the creation of links or **associations** between things.
2. It depends on the ability to recognize **patterns**.
3. It usually requires **repetition** to strengthen those links.

MACHINE LEARNING

Artificial Intelligence (AI) is the field of computer science that focuses on building machines that can perform tasks that usually require human intelligence.

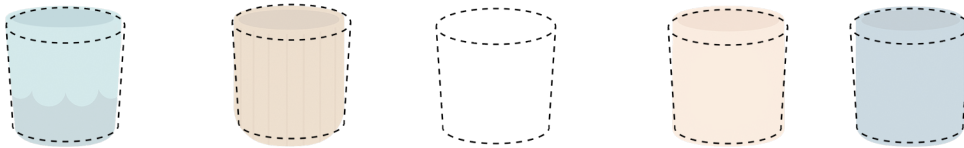
One branch of AI is called machine learning—this is when a computer is designed to have learning capabilities like Maya’s. Through repetition, the computer can identify patterns in data and form internal associations in memory. Once these associations are established and reinforced, the computer can make use of them.

EXAMPLE 1.2

An AI system can also be taught to identify a cup. To do this, it is given an image of a cup and is instructed to associate that image with the word ‘cup’. But to ensure it can recognize cups of all kinds, the computer is shown many images of cups in different sizes, colors, and forms.



By reviewing these images repeatedly, the computer begins to identify the general shape that defines a cup. This ability is called pattern recognition—the capacity to detect recurring regularities, in this case, the overall shape of a cup.



Shared Shape

These images show cups of different sizes, colors, and details. By seeing many different examples, the computer begins to identify the shared shape that makes an object a cup, even when the style, color, or design changes. The ability to identify this shared shape is called pattern recognition.

PATTERN RECOGNITION IS THE ABILITY TO DETECT RECURRING REGULARITIES.

TYPES OF LEARNING

AI systems, like humans, can learn in different ways. There are two main types of machine learning: supervised learning and unsupervised learning.

Supervised learning is like teaching. Maya was given a cup and told, “This is a cup.” This happened repeatedly until she learned to recognize the shape of a cup and associate it with the word ‘cup’.

For an AI system, supervised learning works in a similar way. The computer is given data that already has labels and must find the pattern that links the data to those labels. For example, just as in the previous case, the AI is shown many images of cups, each labeled ‘cup,’ and it learns to associate the recurring shape with the label.

On the other hand, **unsupervised learning** is more like exploring. The AI is not told what to look for. Instead, it analyzes the data, identifies patterns, and groups similar items together based on shared features. Labels are not used during learning, but they can be added afterward to name each group once the patterns are understood.

EXAMPLE 1.3

Imagine Maya walking around the house each day and grouping objects by similarity: cups in one pile, shoes in another, crayons in another. Later, someone tells her the names of these groups: “cups,” “shoes,” and “crayons.” In this way, she would have learned without being told the names in advance.



In unsupervised learning, similar objects are grouped together without being told their names. Later, labels such as ‘cups,’ ‘shoes,’ and ‘crayons’ can be added to each group.

In a similar way, an AI system may be instructed to review hundreds of unlabeled images of household objects and group them based on recurring patterns of shape, color, or design. In such a case, it is said that the AI has learned unsupervised. The labels can be added to those groups afterward.

TWO COMMON TYPES OF LEARNING ARE SUPERVISED LEARNING
AND UNSUPERVISED LEARNING.

BUILDING THE MODEL

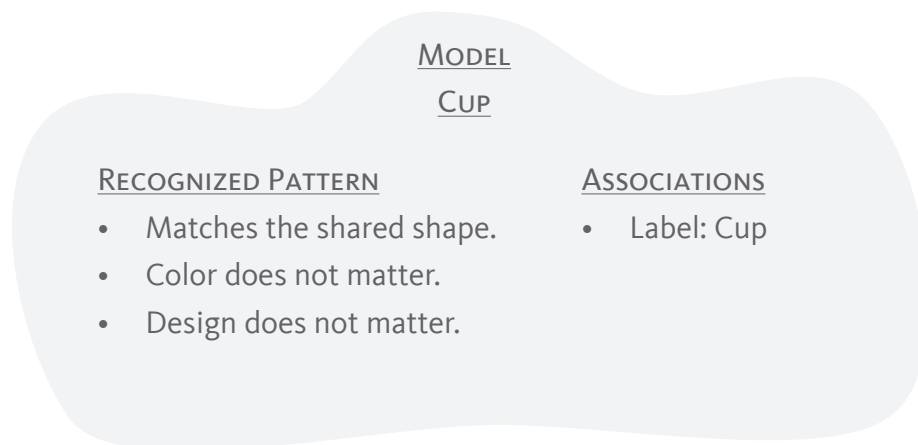
The process of learning results in the creation of a model. A model is the AI's internal representation of what it has learned — made up of the patterns it has recognized, the labels associated with those patterns, and the rules it has formed about which features are important and which are not.

A **MODEL** IS THE INTERNAL SYSTEM THAT REPRESENTS RECOGNIZED PATTERNS AND THE ASSOCIATIONS FORMED FROM THEM.

EXAMPLE 1.4

In this example, we will explore what would be contained in the AI's model of a cup. The most important element is the shared shape—the recurring pattern the AI has learned to identify. But the model also includes refinements, or rules, about features that do not affect whether something is a cup.

For instance, the style, color, or decorative designs are all noted but explicitly



This diagram shows a model of a cup. On the left are the recognized patterns that define a cup. The model identifies the shared shape as essential while explicitly instructing the AI that other variations—such as style, color, or design—do not matter for recognizing a cup. On the right are the associations linked to this pattern, namely the label 'cup'.

treated as irrelevant. As long as the object fits the shared shape, the model recognizes it as a cup.

Finally, the model includes the associations connected to this pattern, such as the label ‘cup’. This enables the computer to generalize, applying what it has learned to correctly identify cups it has never seen before.

TRAINING THE MODEL

Once a model is built, it must be tested to evaluate how well it performs. This testing is done on new data—examples the AI has not seen before. The goal is to determine whether the model can generalize what it has learned and accurately classify new inputs, such as recognizing a cup in an unfamiliar image.

Testing does not happen in isolation. As part of this process, the model’s predictions are compared to the correct answers. When it makes errors—known as misclassifications—those mistakes are analyzed. The model is then refined by adjusting how it interprets patterns and associations.

This ongoing process of learning from mistakes and improving performance by updating the model is known as training.

TRAINING IS THE PROCESS OF IMPROVING A MODEL USING NEW DATA AND CORRECTIONS.

Notes

While there is an official training phase that occurs after a model is first built and before it is released for use, training is not limited to that stage. In reality, training—or more accurately, refinement—is an ongoing process.

EXAMPLE 1.5

Now that the model for a cup has been built, it is tested on new images the AI has not seen before. The AI attempts to classify each image using the model it has created for 'cup'. It checks for the recurring pattern it has learned, disregards irrelevant features like color and design, and predicts which images contain cups based on that pattern.

The model correctly classified some images, but misclassified others. Upon review, it was found that the cups the model missed were those that had handles. The AI was informed that these examples were also cups and had been incorrectly excluded. It was then instructed to review those images, compare the patterns, and refine its model to correct the error.



Correctly Classified



Misclassified

This image shows how the model performed when tested on new data. On the left are cups that the model correctly classified based on the learned pattern. On the right are cups that were misclassified—these were not recognized because the model had not yet learned to accept handles as part of the pattern for 'cup'.

After reanalyzing the images, the model identified a new feature—the handle—and recognized that some cups have handles while others do not. It updated its internal representation accordingly, forming a new rule: “Handles do not matter.”

MODELCUPRECOGNIZED PATTERN

- Matches the shared shape.
- Color does not matter.
- Design does not matter.
- Include handles.

ASSOCIATIONS

- Label: Cup

After training, the model is updated to include cups with handles. It adds a rule to include handles as part of the recognized pattern, allowing the model to classify as ‘cup’ any object that matches the shared shape—whether it has a handle or not.

Now that the initial phase of training is complete, the model is ready for use. But learning does not stop here. While there is an official training phase that takes place after a model is first built and before it is released, training is not limited to that stage.

In practice, learning—or more precisely, refinement—is an ongoing process. As the model encounters new data, makes mistakes, or operates in changing conditions, it can continue to be adjusted and improved.

LEARNING IS A CONTINUOUS PROCESS.

Notes

Remind students: Learning doesn’t stop after the test — it’s a continuous process.

Just like an AI model, we don’t get everything right the first time. We learn through practice, mistakes, feedback, and adjustment. The more we apply what we’ve learned and reflect on our errors, the stronger our internal models become.

SUMMARY

Learning – the ability to form and use new associations.

- The three fundamental attributes of learning:
 1. the ability to recognize patterns.
 2. the ability to form associations.
 3. the need for repetition.
- **Pattern recognition** – the ability to detect recurring regularities.
- Two common types of learning are **supervised** (with labels) and **unsupervised** (without labels).
- **Model** – the internal system that represents recognized patterns and the associations formed from them.
- **Training** – the process of improving a model using new data and corrections.
- Learning is a continuous process.

Activity 1.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>T</u>	Learning is the ability to form and use new associations.
<u>F</u>	Pattern recognition allows a system to detect and respond to random or unrelated features.
<u>T</u>	Supervised learning uses labeled data.
<u>F</u>	Unsupervised learning means the system receives labels in advance and uses them to form associations.
<u>T</u>	A model is the internal system that holds the patterns and associations a computer has learned.
<u>F</u>	Once a model is created, it can no longer be changed or improved.
<u>T</u>	Training is the process of improving a model by using new data and correcting its mistakes.
<u>F</u>	Learning is a one-time event that stops once the initial model is built.
<u>T</u>	A well-trained model can generalize its learning to identify patterns it has never seen before.
<u>T</u>	Repetition helps reinforce associations and improve pattern recognition.

Activity 1.2

Answer the questions below.

1. What is the best definition of learning?
 - a) The ability to remember everything perfectly
 - b) The ability to perform tasks without error
 - c) **The ability to form and use new associations**
 - d) The process of storing all inputs in memory
2. Which of the following is NOT a fundamental attribute of machine learning?
 - a) Recognizing patterns
 - b) Repeating examples
 - c) **Memorizing facts**
 - d) Creating associations
3. What does pattern recognition allow a system to do?
 - a) Respond only to labeled data
 - b) **Detect recurring regularities in inputs**
 - c) Ignore all differences in data
 - d) Randomly sort unrelated data
4. In pattern recognition, which kind of feature is most likely to be ignore?
 - a) The shared structure across examples
 - b) A recurring sound pattern
 - c) The repeated shape of an object
 - d) **An irrelevant design variation**
5. In supervised learning, what is given to the AI system to help it learn?
 - a) Data with missing pieces
 - b) **Data labeled with correct answers**
 - c) Completely random patterns
 - d) Unlabeled examples from multiple sources

 Activity 1.2 (continued)

Answer the questions below.

6. Which statement describes unsupervised learning?
 - a) **The AI finds patterns without labels**
 - b) The AI is told which output is correct
 - c) The AI memorizes all data examples
 - d) The AI ignores unknown patterns

7. What is a model in machine learning?
 - a) A database of all stored inputs
 - b) A system of memorized examples
 - c) A set of outputs from one training session
 - d) **An internal system of learned patterns and associations**

8. What does it mean when a model can generalize?
 - a) It copies exact answers from training data
 - b) It makes up patterns without learning
 - c) It erases data to free memory space
 - d) **It applies learned patterns to new, unseen data**

9. What happens when a model makes a misclassification?
 - a) It automatically deletes the data
 - b) The model is tested on easier data
 - c) **It is retrained or refined to improve**
 - d) The model shuts down for safety

10. Which statement best reflects the idea of continuous learning?
 - a) A model is fixed and never changes after being used
 - b) **Learning continues as the model encounters new data and errors**
 - c) Once trained, a model no longer needs updates
 - d) Learning stops once the system is deployed

Activity 1.3

Read the narrative and answer the questions below.

Plant Learning

A team of scientists is using machine learning to build a system that can identify different types of plants. At first, they collect thousands of plant images and label each one with its name (e.g., “rose,” “sunflower,” “tulip”). They use these labeled examples to teach the AI system what each plant looks like. Over time, the AI begins to recognize important patterns in the shape of the leaves and petals, while ignoring variations like background color or lighting.

Once the model is built, it is tested on a new set of plant images it has never seen before. It correctly identifies many of them but fails to recognize some roses that were photographed from an unusual angle. The scientists examine these mistakes and add more examples showing roses from different angles. The system updates its model and improves its accuracy.

1. What type of learning is being used to teach the AI system to identify plants, and how do you know?

Supervised learning, because the AI system was trained on labeled data (each image had the plant's name).

2. What features did the model learn to focus on, and which did it learn to ignore?

It learned to focus on shapes of leaves and petals (important features) and to ignore background color and lighting (irrelevant features).

3. Why is it important to have the model tested on images it had never seen before?

To check if it could generalize — apply what it learned to new situations.

4. Why did the model fail to recognize some of the roses? What was missing?

The model failed because it hadn't seen enough variety in angle or perspective during training, so it didn't recognize roses in unfamiliar views.

5. How did the model improve, and what process does this describe?

The model improved through training/refinement — it learned from its mistakes and updated its internal representation to become more accurate.

Activity 1.4

Read the narrative and answer the questions below.

Calls of the Wild

Scientists are studying animal sounds recorded in a rainforest. They have thousands of short audio clips, but most of them are not labeled — they don't know which species made each sound. Instead of guessing or labeling by hand, they use unsupervised learning to help them organize the data.

The AI analyzes the audio clips and looks for recurring patterns in pitch, rhythm, and frequency. Over time, it groups sounds that are similar into distinct clusters. Each cluster represents a sound pattern group, but the computer does not know what animal made which sound.

By grouping the data this way, the AI helps scientists discover structure in the sound recordings. This allows researchers to focus their time on analyzing each group and potentially identifying unknown species or behaviors.

1. What kind of learning is the AI using in this scenario, and why?

Unsupervised learning, because the AI is working with unlabeled sound data and grouping it based on similarity.

2. What features of the sounds is the AI analyzing to group them?

The AI analyzes features like pitch, rhythm, and frequency of the sounds.

3. Why doesn't the AI label the sounds during the learning process?

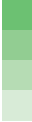
Because it was not given labels — it doesn't know which species made which sound.

4. What does the AI produce at the end — labels or something else?

The AI produces clusters or groups of similar sounds, not labeled categories.

5. How does this help scientists who are studying animals in the wild?

It helps scientists by organizing large amounts of unlabeled data so they can focus on patterns, identify unknown species, or find new behaviors.



Activity 1.5

In this activity, you'll explore how to build, test, and use a machine learning model using an online platform. You'll start by uploading sample images provided to you and then have the option to add your own examples.

Choose a Platform

Start by selecting a simple machine learning platform. Two recommended options are:

- Teachable Machine by Google
- Machine Learning for Kids

Both platforms allow you to upload data, train a model, and test it — but Machine Learning for Kids also supports model refinement and feedback.

Getting Started

- Choose a platform.
- Create a new image project.
- Upload the provided images to build at least two classes (e.g., “cat” and “dog”).
- Click Train Model.
- Test your model using new images.
- (Optional) If the model makes mistakes, add more examples and retrain.

Sample Image Sets

To help you get started, choose one of the following sets. Each contains enough examples to form two distinct classes:

- Cats and Dogs: bit.ly/cats_dogs_ML
- Flowers and Trees: bit.ly/flowers_trees_ML

You may also create your own dataset by taking or drawing pictures and uploading them to the platform.

